QML Lab • research project 14

Noise-Aware Training with Error Mitigation

How much accuracy does zero-noise extrapolation recover for a classifier under realistic noise?

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Quadrant	CQ (classical data, quantum processor)	**Track**	Noise and hardware
Difficulty	Medium	**Tools**	Qiskit Aer, Mitiq, PennyLane
Builds on	Modules 5 and 8 of the QML Lab	**Workspace**	`qml_projects/noise-aware-training-zne/`

Summary. On today's hardware, noise shrinks every expectation value a model produces. This project trains and evaluates the variational classifier of Module 5 under realistic noise models and applies zero-noise extrapolation (ZNE) to recover noise-free values. It separates two questions that are often mixed up: whether noise changes the predicted class, and whether it biases the numbers the model relies on for training and calibration.

1 Background and key concepts

1.1 Noise in variational models

Gate errors, decoherence and readout errors make an expectation value $E(\lambda)$ depend on the noise strength λ, with $\lambda = 1$ for the native device. Under global depolarizing noise, $E(\lambda) \approx (1-p)^d E_0$ shrinks exponentially with depth d. Noise also flattens the cost landscape, a noise-induced barren plateau (Wang et al., 2021). Error mitigation reduces the bias of expectation values without full error correction (Cai et al., 2023).

1.2 Zero-noise extrapolation

ZNE (Temme et al., 2017; Kandala et al., 2019) measures $E(\lambda_k)$ at amplified noise levels $1 = \lambda_1 < \lambda_2 < \dots$ and extrapolates to $\lambda = 0$. Richardson extrapolation uses

$$E(0) \approx \sum_k \gamma_k E(\lambda_k), \qquad \sum_k \gamma_k = 1, \qquad \sum_k \gamma_k \lambda_k^j = 0 \;\; (j = 1, \dots, K-1). \tag{1}$$

Digital ZNE amplifies noise by unitary folding, $G \to G(G^\dagger G)^m$, which leaves the ideal circuit unchanged but multiplies its noise by $\lambda = 1 + 2m$ (Giurgica-Tiron et al., 2020). Mitiq implements these methods for Qiskit and PennyLane circuits (LaRose et al., 2022). The price is variance: extrapolated values need many more shots.

1.3 Why the class may survive noise

If noise only rescales E towards 0, the sign of $f(x) = E(x) + b$ changes only for points close to the decision boundary. Accuracy can therefore be surprisingly robust, while training (which needs gradient magnitudes) and calibrated probabilities suffer. Measuring both is part of the project.

2 Research questions

The project is open-ended: these questions are the starting point, not the finish line. Add new ones to the README and the journal as they come up.

RQ1 Is it better to train with noise, or to train noise-free and mitigate at test time?

RQ2 How much does ZNE recover in accuracy and in expectation-value error, and at what shot overhead?

RQ3 Does a device-based noise model lead to different conclusions than simple depolarizing noise?

3 Method

3.1 Data

Two moons and digits (Project 04 features), with the splits used in Module 5 and Project 04 so that noise-free results are directly comparable.

3.2 Models

The Module 5 re-uploading classifier ($n = 2$–4 qubits, $L = 1$–3 layers). Noise models in Qiskit Aer: depolarizing errors on one- and two-qubit gates ($p_1 = 10^{-3}$, $p_2 = 10^{-2}$) plus readout error, and a noise model built from a real device's calibration data. ZNE with scale factors $\{1, 3, 5\}$ and linear or Richardson extrapolation through Mitiq.

3.3 Baselines

Noise-free training and evaluation (upper bound); noisy evaluation without mitigation; readout-error mitigation alone.

3.4 Metrics

- Test accuracy; mean absolute error of $\langle Z \rangle$ against noise-free values.
- Signal magnitude $\mathbb{E}|\langle Z \rangle|$ as a function of depth and noise.
- Number of circuit executions (shot overhead) per prediction.

4 Planned experiments

Each experiment becomes one `[[experiments]]` entry in the workspace, so its runs, parameters and metrics are stored and compared in the portal.

5 Running it in the QML Lab

The workspace lives in `qml_projects/noise-aware-training-zne/`. Scripts import three helpers from `qml_lab/plots.py`:

ID	Experiment	What varies	Main metrics
E1	Noise sweep	$p_2 \in [0, 0.05]$, no mitigation	accuracy, signal
E2	ZNE	linear vs Richardson, scale factors	error, overhead
E3	Training regimes	train clean/noisy, test noisy $\pm$ ZNE	accuracy
E4	Readout mitigation	on/off	error
E5	Depth	$L = 1..5$ under noise	accuracy, signal

Table 1: Experiment plan. Results go into the journal and the results log (section 9).

- `params()` reads the parameters from the portal's run form (defaults when run by hand);
- `save()` stores figures with the run;
- `log_metrics()` records numbers that appear in the run table.

```
qml_projects/noise-aware-training-zne/
  project.toml          metadata + experiments
  README.md             goal, open questions, ideas
  experiments/          runnable scripts
  docs/main.tex         this document
```

First experiment: This experiment is *planned*: write the script, add the block below to `project.toml`, then run `python manage.py sync_projects`.

```
[[experiments]]
slug = "zne-vqc"
title = "ZNE for the Module 5 classifier under depolarizing noise"
script = "experiments/e01_zne_vqc.py"
figure = "zne_accuracy.png"

[experiments.params]
noise_p1 = 0.001
noise_p2 = 0.01
scale_factors = "1,3,5"
extrapolation = "richardson"
shots = 2000
n_layers = 3
seed = 0
```

```
# from the portal folder: pick up changes to project.toml
python manage.py sync_projects
cd qml_projects/noise-aware-training-zne
# run by hand with the default parameters
PYTHONPATH=../../qml_lab python experiments/e01_zne_vqc.py
```

6 Quality bar

- A classical baseline with a similar number of trainable parameters, tuned with the same care (Bowles et al., 2024).
- At least five random seeds per configuration, reported as mean $\pm$ standard deviation.

- How results change with the number of qubits.
- At least one run under a realistic noise model.
- Report shot overhead next to every mitigated result; mitigation is never free.

7 Open directions

- Probabilistic error cancellation and other mitigation methods (Cai et al., 2023).
- Noise-aware circuit design: fewer two-qubit gates, hardware-native layouts.
- The same study on real hardware (Project 16).

8 Risks and limitations

- ZNE increases variance; small shot budgets can make results worse than unmitigated ones.
- Extrapolation assumes a smooth noise response; non-Markovian device noise can break it.
- Accuracy alone can hide large biases in expectation values.

9 Results log

Summarise findings here as they become stable; day-to-day notes belong in the portal journal.

Date	Run	Finding	Next step

References

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LaRose, R., Mari, A., Kaiser, S., Karalekas, P. J., Alves, A. A. et al. (2022) 'Mitiq: A software package for error mitigation on noisy quantum computers', *Quantum*, 6, 774. Available at: https://doi.org/10.22331/q-2022-08-11-774.

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